INCH-POUND
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SUPERSEDING
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28 May 1996

DETAIL SPECIFICATION

TRANSMISSION, CROSS-DRIVE, CD850 SERIES

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of CD850 series cross-drive transmissions for use in tracked combat vehicles.

1.2 Classification. The transmissions covered by this specification are of the types specified below (see 6.2):

Type I	- CD850-5 for use with gasoline engines.
Type II	- CD850-6 for use with diesel engines.
Type III	- CD850-6A for use with diesel engines.
Type IV	- CD850-6A1 for use with diesel engines.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-2104	- Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service .
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

8348290	- Transmission Assembly, CD850-5, M48A2.
8350025	- Transmission Assembly, CD850-6, M60.
8355830	- Transmission Assembly, CD850-6A, M60A1.
11650111	- Transmission Assembly, CD850-6A1, M60A1.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design, materials, and manufacturing processes. Design, materials and manufacturing process selection shall be as specified herein and in applicable referenced specification, standards, and drawings. Transmission construction, interface, and assembly shall be in accordance with drawings, parts lists, and specifications as shown on Drawings 8348290, 8350025, 8355830, 11650111 and associated documents and publications referenced therein. Unless otherwise specified (see 6.2), the transmission shall be furnished without oil-filler kit, or if not specified, in accordance with the manufacturer's standard practices.

3.2.1 Identification marking. Identification marking shall be permanent and legible and shall include, as a minimum, the manufacturer's name and part number, the part or identifying number (PIN) and the National Stock Number (NSN) (see 6.2).

3.2.2 Service brake adjustment. Unless otherwise specified, the transmission service brakes shall be adjusted so when the brake is applied with 475 pound-foot (lb-ft) [644 Newton meters (n-m)] of torque, the "A" mark on the brake apply cam is aligned with the scribe mark on the anchor to within 0.016 inch (in.) (0.406 millimeter (mm)). After completion of the transmission conformance tests (see 4.3.2), the brake adjustment shall be rechecked and, if necessary, the brakes shall be readjusted to meet the above requirements (see 4.5.4).

3.2.3 Low range and reverse gear band adjustment. The low range and reverse gear bands shall be adjusted for proper running clearance. To adjust the low range and reverse gear clutch bands, the transmission must be stopped. Tighten the band adjusting screw to 30 lb-ft (40 N-m) of torque, then loosen the adjusting screw 300 to 360 degrees (5 to 6 flats on hex head). Tighten the adjusting screw lock nut. Each band must be adjusted separately (see 4.5.5).

3.2.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

3.2.5 Exterior surface treatment. Unless otherwise specified (see 6.2), all exposed surfaces of the transmission shall be cleaned, painted or treated for corrosion resistance as specified on the applicable drawings, or if not specified, in accordance with manufacturers standard procedure.

3.2.6 Oil. The transmission shall be filled to operating level with oil conforming to grade 10 of MIL-PRF-2104 or equivalent when operating in ambient air temperature above 0 degrees Fahrenheit (°F) (-18 degrees Celsius (°C). Oil conforming to MIL-L-46167 or equivalent shall be used when operating in ambient temperature below 0°F. Oil level shall be maintained within the operating range on the oil level indicator.

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3.3 Transmission performance.

3.3.1 Functional operation. The transmission shall operate in all range and steer positions without any indication of mechanical malfunction. In type I transmissions, the speedometer shaft shall rotate in a clockwise direction when the transmission is operated in a forward range. With the transmission not running, the torque effort required to move the selector shaft to any shift position and the torque effort required to move the steer shaft shall be in accordance with table I.

TABLE I. Selector and steer shaft operation.

Position	Type I Torque pound-inch (lb.-in) (N-m)	Type II Torque pound-inch (lb.-in) (N-m)	Type III and IV Torque pound-inch (lb.-in) (N-m)
Selector			
Neutral-low	60 - 100 (6.8-11.3)	60 - 85 (6.8-9.6)	60 - 85 (6.8-9.6)
Low-high	60 - 100 (6.8-11.3)	60 - 85 (6.8-9.6)	60 - 85 (6.8-9.6)
High-reverse	60 - 100 (6.8-11.3)	60 - 85 (6.8-9.6)	60 - 85 (6.8-9.6)
Steer			
Neutral-full left	30 max. (3.39)	30 max. (3.39)	40 - 70 (4.5-7.9)
Neutral-full right	30 max. (3.39)	30 max. (3.39)	40 - 70 (4.5-7.9)

3.3.2 High range stall. When tested as specified in 4.5.7.1,the transmission high range stall performance shall be in accordance with table II.

TABLE II. High range stall performance.

Item	Type I	Type II and III	Type IV
Input torque (lb-ft) (N-m)	650 (881)	750 (1016)	750 (1016)
Input speed revolutions per minute (rpm)	See figure 1	See figure 2	1300 - 1400
Output torque (lb-ft) (N-m)	See figure 1	See figure 2	2840 min. (3850)
Oil pressure pounds per square inch (psi) (kilopascals (kPa))			
Main	108 - 152 (744 - 1048)	107 - 143 737 - 985)	115 - 143 (792 - 985)
Lube	7 min. (48)	3 min. (21)	10 min. (69)
Oil flow main gallons per minute (gal/min) (liters per minute L/min.))	23 min. (87)	19 min. (34)	22 min. (83)
Oil cooler (gal/min) (L/min.)			28 min. (106)

3.3.3 High range no-load. When tested as specified in 4.5.7.2, the transmission high range no-load performance shall be in accordance with table III.

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TABLE III. High range no-load performance.

Item	Type I	Type II and III	Type IV
Input torque (lb-ft) (N-m)	135 max. (183)	180 max. (244)	180 max. (244)
Output speed (rpm)	1360 min.	1585 min.	1585 min.
Oil pressure (psi) (kPa)			
Main	113 - 169	113 - 169	122 - 146
	(779 - 1165)	(779 - 1165)	(779 - 1165)
Converter “in”	70 min.	80 min.	95 min.
Lube	40 - 70	42 - 71	49 - 71
Oil flows (gal/min)			
Main	47 min.	45 min.	52 - 61 min.
Converter “out”	37 min.	31 min.	36 min.

3.3.4 Low range stall. When tested as specified in 4.5.8.1, the transmission low range stall performance shall be in accordance with table IV.

TABLE IV. Low range stall performance.

Item	Type I	Type II, III and IV
Input torque (lb-ft) (N-m)	648 to 652 (878 - 883)	748 to 752 (1014 - 1019)
Input speed (rpm)	1855 max.	1410 max.

3.3.5 Low range no-load. When tested as specified in 4.5.8.2 the transmission low range no load performance shall be in accordance with table V.

TABLE V. Low range no-load performance.

Item	Type I	Type II and III	Type IV
Input torque (lb-ft) (N-m)	152 max.	161 max.	145 max.
Output speed (rpm)	492 min.	573 min.	573 min.
Oil pressure (psi) (kPa)			
Main	184	179 min.	200 min.
	(1268 min.)	(1234 min.)	(1379 min.)
Converter “in”	63-95	65-95	75-100
Lube	26 min.	26 min.	40 min.

3.3.6 Reverse range stall. When tested as specified in 4.5.9 the transmission reverse range stall performance shall be in accordance with table VI.

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TABLE VI. Reverse range stall performance.

Item	Type I	Type II, III and IV
Input torque (lb-ft) (N-m)	648 to 652 (878 to 883)	748 to 752 1013 to 1018)
Input speed (rpm)	1855 max.	1410 max.
Oil pressure (psi (kPa)		
Main	173 min. (1193 min.)	173 min. (1193 min.)

3.3.7 Neutral steer. When tested as specified in 4.5.10 the right and left steer clutch pressure must increase gradually as the steer control is shift is moved, and at full right and left steer the transmission performance shall be in accordance with table VII.

TABLE VII. Full steer performance.

Item	Full L.H. steer 1/	Full R.H. steer 1/
Oil pressure (psi) (kPa)		
R.H. steer clutch	-	178 min. (1227 min.)
L.H. steer clutch	178 min. (1227 min.)	-
Lube	7 min. (48 min.)	7 min. (48 min)

1/ L.H. and R.H. are defined in 6.3.

3.3.8 Service brake capacity. When tested as specified in 4.5.11. The right and left hand brakes shall hold the right and left hand outputs without movement.

3.3.9 Leakage. During and after an operational simulation, the transmission shall show no evidence of a leak or drip (see “c” and “d” of 6.4).

3.3.10 Durability. The transmission shall require no repair and no maintenance other than filter changes, oil changes, brake adjustments and low range and reverse band adjustments during vehicle operation at a proving ground for distances up to 5000 miles (8047 kilometers (km).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
   1. Preproduction inspection (see 4.2.1)
   2. Initial production inspection (see 4.2.2)

b. Conformance inspections (CI) (see 4.3)
   1. Examination (see 4.3.1)
   2. Tests (see 4.3.2)

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4.2 First article inspection. First article inspection shall be performed on preproduction or initial production units as specified when a first article is required. Approval of the first production units by the government shall not relieve the contractor of the obligation to supply transmission that are fully representative of those inspected as a first article (see 6.2 & 6.7).

4.2.1 Preproduction inspection. When specified, (see 6.2), the preproduction sample shall consist of one unit. Preproduction inspection shall consist of exams and tests as specified in Table VIII.

4.2.2 Initial production inspection. Unless otherwise specified, (see 6.2), one of the first production transmissions and component, subassemblies installed in the vehicle stall be subject to the durability test.

TABLE VIII. Verification methods.

Title	Requirement	Inspection	First article inspection		Conformance Inspection	
			Prepro-duction	Initial	Exam	Test (100%)
Design, materials, and manufacturing processes	3.2	4.5.3	X		X	
Identification marking	3.2.1	4.5.3	X	-	X	
Service brake adjustment	3.2.2	4.5.4	X	-		X
Low range and reverse gear band adjustment	3.2.3	4.5.5	X	-		X
Exterior surface treatment	3.2.5	4.5.3	X		X	
Oil	3.2.6	4.5.12 & 4.5.12.1	X	-		X
Functional operation	3.3.1	4.5.6	X	-		X
High range stall	3.3.2	4.5.7.1	X	-		X
High range no-load	3.3.3	4.5.7.2	X	-		X
Low range stall	3.3.4	4.5.8.1	X	-		X
Low range no-load	3.3.5	4.5.8.2	X	-		X
Reverse range stall	3.3.6	4.5.9		-		X
Neutral steer	3.3.7	4.5.10	X	-		X
Service brake capacity	3.3.8	4.5.11.1 & 4.5.11.2	X	-		X
Leakage	3.3.9	4.5.12	X	-		X
Durability	3.3.10	4.5.13	-	X		-

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4.3 Conformance inspection. Conformance testing shall include the examinations and the tests as specified in table VIII. Noncompliance with any of the specified requirements shall be cause for rejection (see 6.8).

4.3.1 Examination. Each transmission shall be examined for defects as specified in table IX.

4.3.2 Tests (100% inspection). Each transmission shall be subject to the conformance inspection tests specified in table VIII.

4.4 Test conditions. Testing shall be performed with transmission installed on any appropriate test stand. Provisions shall be made for connecting input and output shafts, range selector, steering and brake controls, pressure and temperature outlets, and oil lines. Test shall be conducted at room temperature. Equipment, instrumentation, and conditions shall be provided as follows:

a. Input power supply of approximately 850 lb-ft (1152 N-m) torque at 1000 to 2500 rpm.
b. Stall devices for output shafts (10 000 lb-ft torque) (13 558 N-m).
c. Levers for operating range selector, steering controls, and brake controls.
d. Oil supply system that includes oil lines, cooler, filters, and connections for pressure and temperature outlets.
e. Instrumentation for measuring the following:
    (1) Input torque in lb-ft (N-m).
    (2) Input speed in rpm.
    (3) Output torque in lb-ft (N-m).
    (4) Main oil pressure in psi (kPa).
    (5) Lubricant oil pressure in psi (kPa).
    (6) Converter “in” oil pressure in psi (kPa).
    (7) Steer clutch apply oil-pressure in psi (kPa).
    (8) Main oil flow in gal/min (see NOTE).
    (9) Cooler oil flow in gal/min.
    (10) Selector and steer shafts torque in lb.-in. (N-m).
    (11) Brake apply shaft torque in lb-ft (N-m).

NOTE: Main oil flow may be measured by removal of the oil filter assembly and installation of a locally fabricated flow meter adapter. An external oil filter of equal efficiency shall then be used.

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Table IX Classification of defects.

Category	Defects	Method of examination
Critical	None	
Major		
101	Dimensions affecting interchangeability, out of tolerance (see 3.2).	SIE 1/and Visual
Minor		
201	Dimensions affecting interchangeability (see 3.2).	SIE 1/and Visual
202	Identification marking, improper or not legible (see 3.2.1).	SIE 1/and Visual
203	Paint- finish coat, improper application or color (see 3.2.5).	Visual

1/ SIE = Standard Inspection Equipment.

4.5 Verification methods. Acceptable verification methods included in this section are visual inspection, and measurement, sample tests, full-scale demonstration tests, simulation, modeling engineering evaluation, component properties analysis, and similarity to previously-approved or previous-qualified designs.

4.5.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.5.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ±18°F (23 ±10°C).
b. Barometric pressure: 28.5 (+2, - 3) in. [725 (+50, -75) mm) of mercury (Hg).
c. Relative humidity: 50 ±30 percent.

4.5.3 Design, materials, and manufacturing processes. Conformance to 3.2, 3.2.1, & 3.2.5 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawing, specification, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards test reports, and rating data.

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4.5.4 Service brake adjustment check. The brakes shall be adjusted as specified and checked for meeting performance requirements. Each brake shall be adjusted separately with the plugs removed from the brake inspection holes.

4.5.5 Low range and reverse gear band adjustment check To determine conformance to 3.2.3, the low range and reverse gear bands shall be adjusted as specified and checked for meeting performance requirements. Each band shall be adjusted separately.

4.5.6 Functional test. The range selector shall be placed in high range. With the output shafts stalled, increase the input speed to 980 to 100 rpm and run at this speed for a minimum of one minute. With the output shafts coupled to the dynamometers but with no load on the output shafts, run at 980 to 1000 rpm, and functionally check all steer and shift positions for any indication of mechanical malfunction.

4.5.7 High range tests. The transmission shall be operated, as specified in 4.5.7.1 and 4.5.7.2, and checked for meeting the specified performance requirements.

4.5.7.1 Stall. To determine conformance to 3.3.2, the transmission shall be operated with the range selector in high range, the output shafts stalled and the converter “out” oil temperature stabilized at 200 to 225°F (93 to 107°C). If the transmission stall performance does not fall within the parameters in table II, transmission has failed the test.

4.5.7.2 No-load. To determine conformance to 3.3.3, the transmission shall be operated with the range selector in high range but no load on the output shafts and with 1790 to 1810 rpm input speed, and the converter “out” oil temperature stabilized at 180 to 225°F (82 to 107°C). If the transmission performance does not fall within the parameters in table III, transmission has failed the test.

4.5.8 Low range tests. The transmission shall be operated, as specified in 4.5.8.1 and 4.5.8.2, and checked for meeting the specified performance requirements.

4.5.8.1 Stall. To determine conformance to 3.3.4, the transmission shall be operated with the range selector in low range, the transmission service brakes applied to stall output shafts, and the converter “out” oil temperature stabilized at 200°F (93°C). If the transmission stall performance does not fall within the parameters in table IV, transmission has failed the test.

4.5.8.2 No-load tests. To determine conformance to 3.3.5, the transmission shall be operated with the range selector in low range but no load on the output shafts, the transmission service brakes released, and with 1790 to 1810 rpm input speed and the converter “out” oil temperature stabilized at 180 to 225°F (82 to 107°C). If the transmission performance does not fall within the parameters in table V, transmission has failed the test.

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4.5.9 Reverse range stall test. To determine conformance to 3.3.6, the transmission shall be operated with the range selector in reverse range, the transmission service brakes applied to stall the output shafts, and the converter "out" oil temperature stabilized at 200 to 225°F (93 to 107°C), and checked for meeting the specified performance requirements. If the transmission stall performance does not fall within the parameters in table VI, transmission has failed the test.

4.5.10 Steer test. To determine conformance to 3.3.7, the transmission shall be operated with the range selector in neutral but no load on the output shafts, and with 1790 to 1810 rpm input speed, and the converter "out" oil temperature stabilized at 180 to 225°F (82 to 107°C). The lube oil pressure must be 20 psi minimum at neutral steer position. Rotate the steer control shaft to apply full right left steer. The transmission shall be checked for meeting the specific performance requirement at both full right and left steer. If the transmission stall performance does not fall within the parameters in table VII, transmission has failed the test.

4.5.11 Service brake capacity test.

4.5.11.1 Type I. With the range selector in low range but no load on the output shafts, apply 475 lb-ft (644 N-m) torque on the right hand brake apply shaft and 235 lb-ft (318 N-m) torque on the left hand brake apply shaft. Increase the input speed to 1635 rpm. The above requirements shall be repeated for the right hand brake by applying 475 lb-ft torque on the left hand brake apply shaft and 235 lb-ft torque on the right hand brake apply shaft. The right and left hand brake shall hold the right and left hand outputs without movement.

4.5.11.2 Types II, III, IV. With the range selector in low range but no load on the output shafts apply 475 lb-ft (644 N-m) torque on the right hand brake apply shaft and 180 lb-ft (240 N-m) torque on the left hand brake apply shaft. Increase the input speed to 1260 rpm. The above requirements shall be repeated for the right hand brake by applying 475 lb-ft torque on the left hand brake apply shaft and 180 lb-ft torque on the right hand brake apply shaft. The right and left hand brake shall hold the right and left hand outputs without movement.

4.5.12 Oil leakage check. The transmission shall be operated at an input speed of 1000 rpm but no load on the output shafts, and the service brakes released. The converter "out" oil temperature shall be 180 to 250°F (82 to 121°C) while the transmission is shifted into all range and steer positions. During and after operations the transmission shall be checked for leakage of lubricants (see 6.6).

4.5.12.1 Oil, oil level and temperature, and leakage check. Oil designated as grade 1 of MIL-L-21260 or equivalent shall be used and maintained at operating level during the transmission evaluation and production surveillance runs. Unless otherwise specified, see (6.2) the cooler "in" oil temperature shall be maintained between 160 and 250°F (71 and 121°C) during evaluation. During the evaluation, the oil temperature shall be not more than 300°F (149°C) and there shall be no evidence of leakage. After vehicle has been stopped for three minutes the transmission, while running in neutral range, shall be checked for oil level, oil temperature, and leakage of oil.

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4.5.13 Transmission durability. The transmission and component subassemblies, installed in the vehicle, shall be operated for distances up to 5000 miles (8047 kilometers (km). Operation shall include all types of terrain conditions and ambient temperatures which the vehicle is required to be subjected to in accordance with the provisions of the specification covering the vehicle.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The transmissions covered by this specification are intended for use in tracked combat vehicles and are considered military unique because they must be able to operate satisfactorily at ambient temperatures ranging from -65°F to +300°F. Commercial components are not designed to withstand such extreme environmental condition and would experience failure.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of transmission required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
d. When first article is required (see 3.1 & 4.2).
e. Oil filler kit, if required (see 3.2).
f. PIN and quantity required (see 3.2.1 and 6.5).
g. If identification marking is other than as specified .(see 3.2.1).
h. If exterior surface treatment is other than as specified (see 3.2.5).
i. If preproduction inspection, is required (see 4.2.1).
j. If initial production inspection is not required (see 4.2.2).
k. If inspection conditions are other than as specified (see 4.5).
l. If cooler "in" oil temperature is to be other than as specified (see 4.5.12.1).
m. Packaging requirements (see 5.1).

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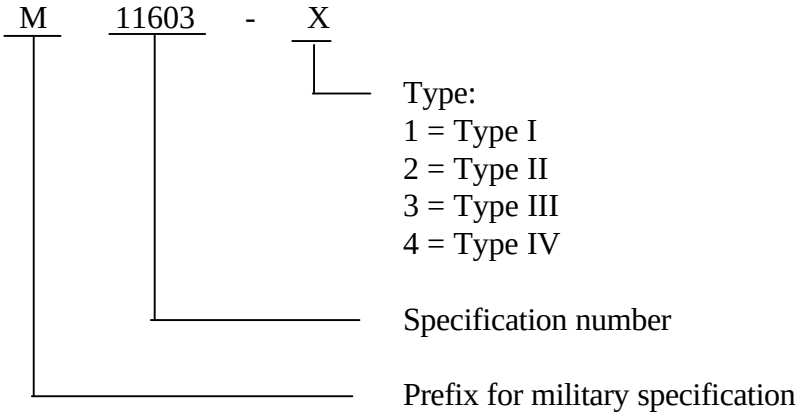
6.3 Transmission orientation. The terms L.H. and R.H. are derived and defined as follows:

a. The "front" of the transmission is the side which mounts to the engine.
b. The "rear" of the transmission is the side opposite the front.
c. The "right" side is the side on the viewer's right as he faces the transmission from the rear.
d. The "left" side is the side on the viewer's left as he faces the transmission from the rear.
e. These definitions are established relative to the transmission only and not relative to any vehicle installation definitions.

6.4 Definitions. The following definitions are established as guidelines governing leakage (see 3.3.8):

a. Weep - Any evidence of fluid beyond a seal or splitline.
b. Seep - Any evidence of fluid beyond a splitline that does not result in the formation of a droplet.
c. Leak - Any evidence of fluid beyond a seal or splitline that results in the formation of a droplet.
d. Drip - Any evidence of fluid beyond a seal or splitline where droplets form and fall.

6.5 Part identification number (PIN). The PINs to be used for transmission acquired to this specification are created as follows:



6.6 Oil leakage. The transmission oil filter should be cleaned. The pressure gage connection fittings should be removed and replaced with specified pipe plugs. All exterior surfaces of the transmission should be cleaned thoroughly (see 4.5.12).

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6.7 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production 0testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

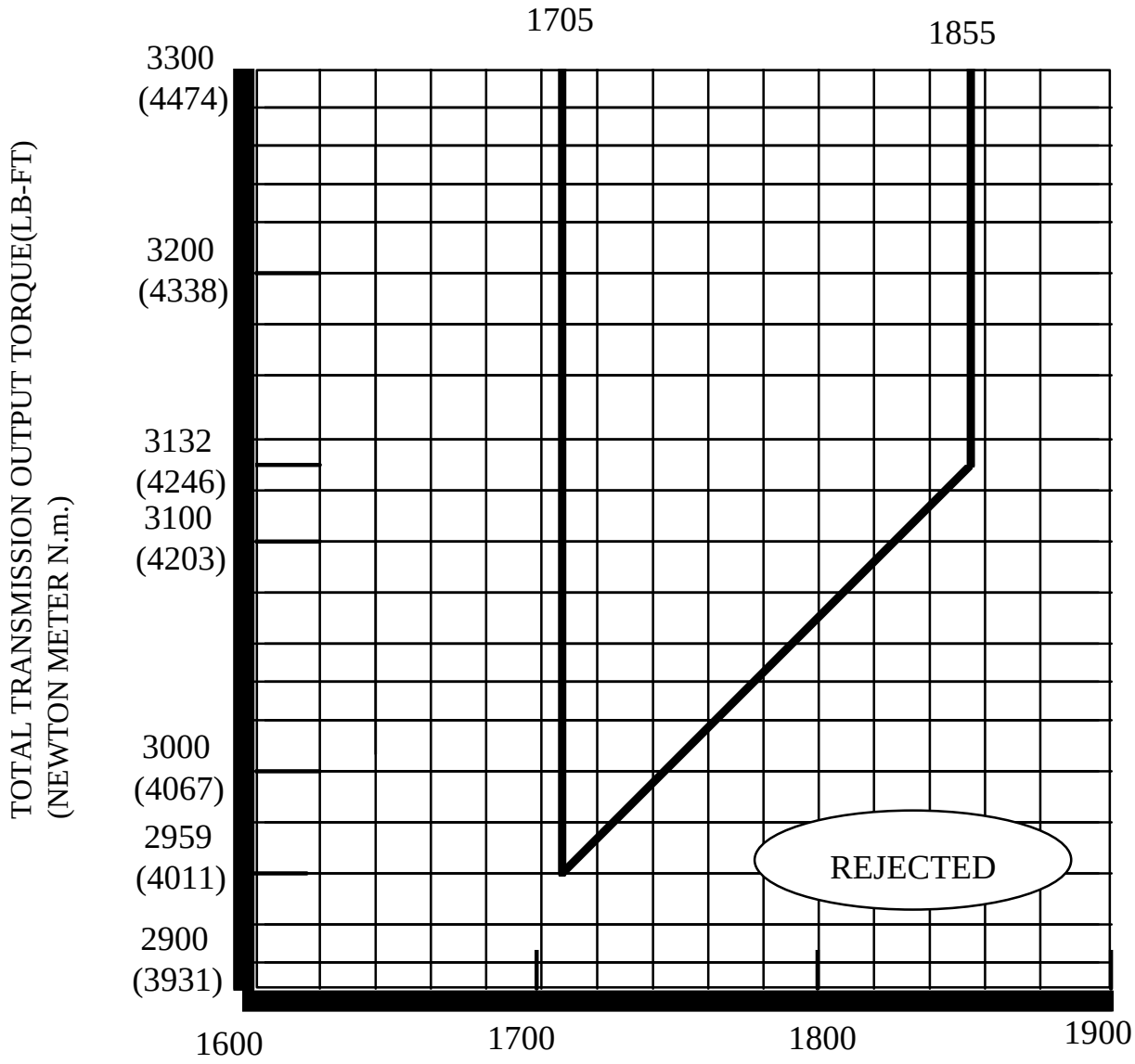
6.8 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table II and their designated frequency based on a risk assessment for the procurement.

6.10 Subject term (key word) listing.

Gear
Gear box
Military combat vehicles

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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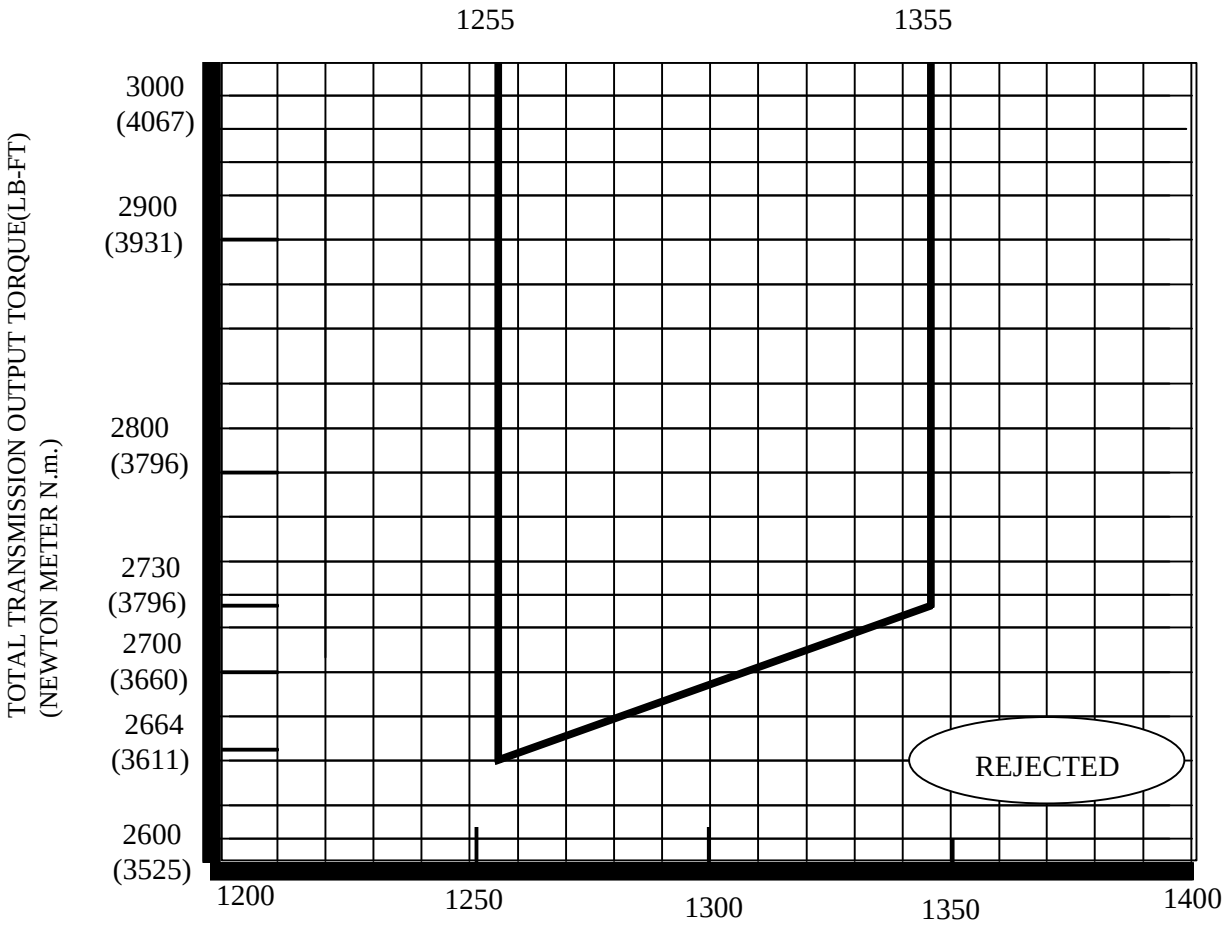


NOTES:

1. Converter out oil temperature 200 to 225°F (93 to 107°C).
2. All figures shown in parenthesis are measured values expressed in metric terms.

FIGURE 1. CD 850-5 transmission acceptance limits high gear stall.

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TRANSMISSION INPUT SPEED (RPM) AT 750 LB-FT (1016 N-M) INPUT TORQUE

NOTES:

1. Converter out oil temperature 200 to 225°F (93 to 107°C).
2. All figures shown in parenthesis are measured values expressed in metric terms.

FIGURE 2. CD-850-6 and CD-850-6A transmission acceptance limits high gear stall.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2520-0020)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-11603G(AT)	2. DOCUMENT DATE *(YYMMDD)* 980904

3. DOCUMENT TITLE
Transmission, Cross-Drive, CD850 Series

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON (810) 574-8745 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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